

AKASH S

MACHINE LEARNING ENGINEER

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Machine Learning Engineer with strong command of **deep learning, NLP and computer vision**, building and deploying models end-to-end in Python with PyTorch, TensorFlow and scikit-learn. Experienced across CNNs, RNN/GRUs, Transformers, GANs and diffusion models, with hands-on work in predictive modeling, anomaly detection, model evaluation and explainability (SHAP/LIME), and serving models as scalable FastAPI services.

EXPERIENCE

Digital Blinc — Machine Learning Intern

Remote, Jun 2025

- Developed an NLP sentiment-analysis model classifying customer feedback at **85% accuracy**; performed EDA to identify model-driving features and trends.
- Engineered text preprocessing and feature pipelines (tokenization, TF-IDF) and evaluated models with precision, recall and F1.

Jyesta Corporate Ltd — Machine Learning Intern

Remote, Jan-Mar 2024

- Built a predictive-maintenance model with Random Forest, reducing simulated equipment downtime by **15%**.
- Optimized data pipelines to cut training latency **20%** via feature vectorization; added Isolation-Forest anomaly detection for fraud.

Hackwise 2.0 — 1st Prize Winner

KVG College of Engineering, Sullia, Karnataka, Apr 2026

- Won a national AI hackathon among **1,000+ teams**; delivered LLM-driven skill-vetting and recommendation models for an AI marketplace.
- Built weighted recommendation scoring and automated AI quiz generation to rank and vet candidate freelancers.

PROJECTS

NeuroSense AI — Emotion Model

PyTorch · Transformers · SHAP

- Designed a cross-modal transformer (4-head, 2-layer) fusing Wav2Vec2 speech, ViT facial and RoBERTa text models with GRU temporal tracking.
- Achieved real-time inference at **<600ms** with ONNX/FP16 optimization and full SHAP/LIME attribution per prediction.

RadiSynx — Medical Image Synthesis

U-Net · GAN · Diffusion

- Built an MRI→CT synthesis pipeline: Attention Residual U-Net → Pix2Pix GAN → DDPM diffusion enhancement on DICOM/NIFTI volumes.
- Evaluated outputs with SSIM, PSNR, MAE and Dice; added Grad-CAM explainability and real-time WebSocket slice streaming.

AI Finance — Forecasting & Anomaly Detection

Prophet · PyOD · sklearn

- Built time-series spend forecasting (Prophet), PyOD fraud/anomaly detection and a 95%+ accurate transaction categorizer.
- Containerized the ML service with Docker and exposed predictions through versioned FastAPI endpoints.

TECHNICAL SKILLS

Languages: Python, Java, SQL, C++

ML / DL: PyTorch, TensorFlow, Keras, scikit-learn, HuggingFace Transformers

Models: CNN, RNN, GRU, Transformers, GANs, Diffusion, XGBoost, Random Forest

NLP & CV: NLP, NLTK, spaCy, Computer Vision, OpenCV, YOLO, rPPG

MLOps & Data: Prophet, PyOD, SHAP, LIME, EDA, Pandas, NumPy

Serving: FastAPI, Flask, Docker, Kubernetes, MLOps, Streamlit

Databases: MySQL, MongoDB, PostgreSQL, Git

Soft Skills: Problem Solving, Analytical Thinking, Experimentation, Communication

LANGUAGES

English (Professional) | Tamil (Native) | Hindi (Conversational)

EDUCATION

PSNA College of Engineering & Technology

B.E. Computer Science & Engineering | Dindigul

2023 - 2027 | CGPA: 8.32

Coursework: DSA, OOP, DBMS, Statistics

Lakshmi School, Madurai

12th: 89.20% | 10th: 92.33%

CERTIFICATIONS

- Microsoft — Generative AI Fundamentals
- AWS — Generative AI Foundations
- UiPath — Agentic Automation Professional
- NPTEL — Joy of Computing in Python

ACHIEVEMENTS

- 1st Place — Hackwise 2.0, 1,000+ teams (2026)
- Top 5 Finalist — Codethon, Coimbatore (2026)
- Academic Excellence - CGPA 8.32 / 10